

# Automated Timesheet Data Migration Using SSIS Packages

## Project Overview:

Graduates will design and implement a scalable **SQL Server Integration Services (SSIS)** solution to **migrate timesheet data from Excel files into a relational database**. The solution must follow **best database design principles** while ensuring data integrity, observability, and automation throughout the process.

## Project Scope:

Participants will focus on the following key tasks:

- **Data Extraction & Looping:**

- Develop SSIS packages that **loop through multiple folders** containing various Excel files.
- Dynamically extract data from different sheets within Excel workbooks.

- **Database Design & Implementation:**

- Design and implement **normalized database tables** for efficient storage and retrieval.
- Ensure adherence to **data integrity constraints** and indexing best practices.

- **Data Processing, Auditing & Observability:**

- Implement logic to **track inserts, updates, and deletions** during data migration.
- Create a **centralized audit log** capturing all transactional changes.
- Utilize **Elastic Stack (Elasticsearch, Logstash, Kibana)** to send audit logs and metrics for real-time monitoring, visualization, and anomaly detection.

- **Automation & Monitoring:**

- Configure **alerting mechanisms, scheduling, SQL Server jobs, and triggers** to streamline execution and data validation.
- Enable real-time monitoring through **Kibana dashboards** and Elastic Stack visualization.

- **Deployment Automation:**

- Implement **GitHub Actions** for automating SSIS package deployment.
- Ensure **version control** and **continuous integration** practices for maintaining code consistency.

## Expected Deliverables:

- **Fully functional SSIS package** with documentation.
- **Optimized database schema design** aligned with best practices.
- **Audit logs and observability dashboards** for tracking execution and detecting anomalies.

- **Automated deployment pipeline** integrated with GitHub Actions.

**Learning Outcomes:**

Graduates will gain hands-on experience with:

- SSIS package development and **data transformation techniques**.
- **Database design principles**, indexing, and normalization.
- **Monitoring and observability** using Elastic Stack.
- **Alerting, scheduling, and automation** for data workflows.
- **GitHub Actions for continuous deployment** of SSIS solutions.

This project will equip graduates with **real-world experience** in data migration, **observability strategies**, and **automation workflows**, making them better prepared for industry demands.